Laboratory 11

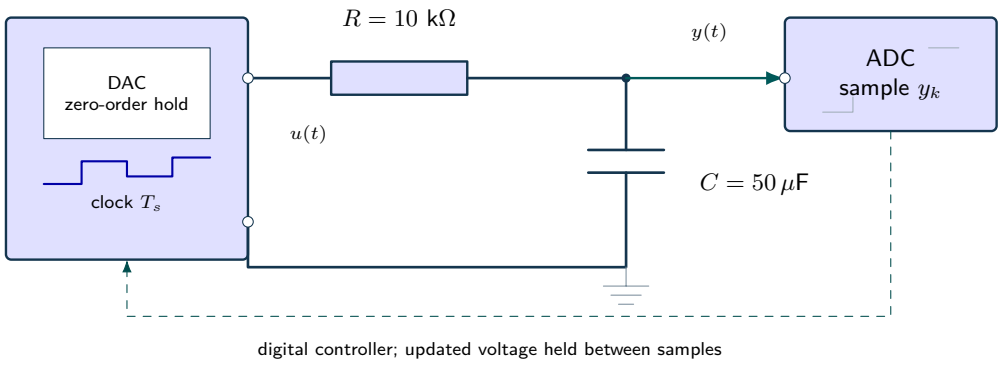
Sampling can destabilize a continuous-time design

Exercise sheet – five integrated problems

Evidence boundary. Original hypothetical scenarios. Derive, calculate and justify decisions; every numerical record is synthetic. Figures are conceptual and not to scale. **Shared nominal data.** Plant: $\dot{y} = -2y + 2u$. Nominal digital PI applies $u_k = 2(r_k - y_k) + I_k$ before updating $I_{k+1} = I_k + 4T_s(r_k - y_k)$. The exact hold and controller update order are part of the model.

1. A sampled controller connected to a continuous RC circuit

The physical example is an ideal 10 kΩ resistor and 50 μF capacitor, with held input voltage u and capacitor voltage y. An ideal ADC/DAC pair connects the circuit to a digital PI controller.



Continuous energy store
$RC = 0.5$ s; $\dot{y} = 2(u - y)$.
The held voltage drives a real-valued continuous state between samples.

Clocked interface
$y_k = y(kT_s)$.
$u(t) = u_k$ for $kT_s \le t < (k+1)T_s$.
Clock and update order matter.

Ideal converter boundary
No quantization, electrical loading or converter dynamics. Use exact held-input discretization when comparing sample periods.

A held voltage drives an ideal RC network. The capacitor voltage is sampled by an ideal ADC; the software controller updates the DAC once per period. Quantization, converter lag and electrical loading are omitted.

(a) The plant is $\dot{y} = -2y + 2u$. Derive its exact update for input held constant over a sampling interval T_s, and evaluate coefficients at $T_s = 0.5$ s.

(b) Use $u_k = 2(r_k - y_k) + I_k$ and $I_{k+1} = I_k + 4T_s(r_k - y_k)$. Derive the state update matrix for $[y_k, I_k]^T$ and its reference-input vector.

(c) Derive the continuous-time PI characteristic polynomial for $k_p = 2$, $k_i = 4$ and compare its poles with what a sampled implementation must check separately.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

2. A stability proof instead of a sampling-rate guess

Derive the exact sampled characteristic polynomial and apply all second-order Jury conditions. Keep the strictness of the interval explicit.

(a) Derive the characteristic polynomial $z^2 - tz + d$ of the update matrix. Express t and d using $a = e^{-2T_s}$.

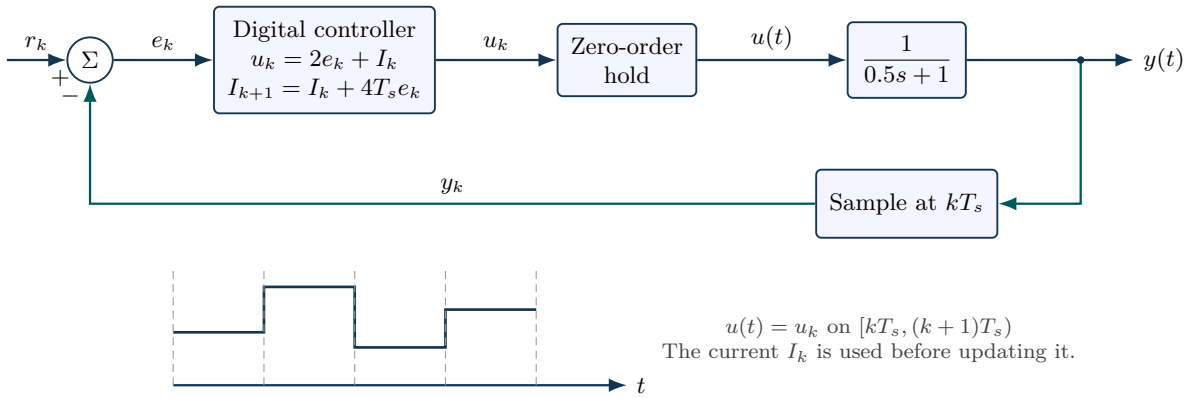
(b) Use the real second-order Jury conditions to prove strict stability for $0 < T_s < 0.75$ and failure for $T_s \geq 0.75$. Show the three inequalities rather than only checking selected samples.

(c) Retain $k_p = 2$ but allow positive k_i. Derive a necessary sampling bound from the determinant condition alone. Evaluate it at $k_i = 8$ and state why the other Jury inequalities are still needed.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

3. The onset of sustained sampled oscillations

Compare stable, marginal and unstable sample periods. Interpret the boundary in physical time, not only as a point in the discrete z-plane.



The sampler, digital update order and hold are explicit. Continuous plant dynamics and sampled controller states occupy different parts of the loop.

(a) At $T_s = 0.75$, find determinant, trace and the nature of the poles. Explain why this endpoint is excluded even though no pole magnitude exceeds one.

(b) Compute the spectral radius for $T_s = 0.05$, 0.5 and 1 s. Classify each implementation and relate the result to the analytic interval.

(c) At $T_s = 0.75$ s, derive the oscillation angle per sample and corresponding angular frequency per second from the conjugate unit-circle pair.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

4. An equilibrium is not a convergence guarantee

The sampled controller has a nominal reference equilibrium even at sample periods that make it unstable. Separate equilibrium equations from local behavior near that equilibrium.

(a) For a mode with discrete magnitude ρ, define an equivalent envelope decay rate per second. Compare the two stable sampling periods in Problem 3(b) and explain why this comparison still does not fully describe their transients.

(b) For constant reference r and any positive sampling interval, derive the nominal equilibrium of the discrete update. Explain why existence of this equilibrium does not ensure convergence to it.

(c) At $T_s = 1$ s and constant r, initialize exactly at $[y, I] = [r, r]$. Explain why a perfect simulation can remain there forever although the implementation is unstable, and design a revealing numerical test.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

5. Implementation order and what happens between samples

Two programs can implement the same displayed PI gains but update the integrator in different orders. Compare the resulting matrices and reconstruct the actual continuous circuit motion.

(a) Now update $I_{k+1} = I_k + 4T_s(r_k - y_k)$ first and apply $u_k = 2(r_k - y_k) + I_{k+1}$ during the next plant interval. Derive the new matrix and show directly that $T_s = 0.5$ is unstable.

(b) Given y_k and held input u_k, derive $y(kT_s + \tau)$ for $0 \le \tau \le T_s$. Explain why a plot of sample values alone may conceal an implementation error or performance issue.

(c) For this first-order circuit under a truly held input, prove whether an intersample extremum can exceed both endpoint values. Explain why that result does not generalize automatically to a higher-order plant.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.